TRƯỜNG ĐẠI HỌC BÁCH KHOA HÀ NỘI **TRƯỜNG ĐIỆN – ĐIỆN TỬ** **Number: 01** **No. pages: 5**	**MIDTERM EXAM 2022.1** **COURSE ID: ET3300E – DIGITAL SYSTEM DESIGN 2** **ELECTRONIC & TELECOM ENGINEERING** **Date: 29/12/2022** *(Books are allowed, PCs are allowed without using keyboard. Return these sheets with answers filled)*
Signatures Instructor:	Group leader:
Student name:	**Student ID:**
Total points: 240	**You get: /240**

Note that:

- For questions with required explanation, the right choice(s) of your answer is only made 40% of points. The rest of 60% of points is made by details you provide.

PART A: QUESTIONS FOR THE INTEL MICROPROCESSORS				
Question 1. What is the instruction moving data from a memory location to a register?			**Point: 10**	**Get:**
A. MOV [4321h], AL	B. MOV CH, [DI]	C. MOV AH, [SI+21h]	D. MOV AL, 4321h	
For your choice(s), please give brief explanation for the data size, and what is the physical address of memory location? …… …… ……				
Question 2. What is the instruction moving data from a register to a memory location?			**Point: 10**	**Get:**
A. MOV [DI+43h], CX	B. MOV AX, [SI]	C. MOV [SI+21h], DX	D. MOV DH, [1234h]	
For your choice(s), please give brief explanation the data size, and physical address of memory location? …… …… ……				

Question 3. What is the meaning/activity of the following instructions in the 8086?		**Point: 50**	**Get:**
No.	**Instruction**	**Detailed Activity/meaning**	
1.	MOV BYTE PTR [DI+21h], [SI+21h]		
2.	MOV CX, [SI+21h]		
3.	PUSH DX		
4.	POP AX		
5.	JMP BX		
6.	JNZ LabelA		
7.	MOVSW		
8.	ROL DL,04		
9.	IN AL, 00h		
10.	OUT DX, AL		
11.	TEST DL, CL		
12.	CMP BH,20		
13.	SHL CL,10		
14.	DIV CH		
15.	MUL DL		
16.	IDIV BX		
17.	IMUL BL		
18.	LOOP NEXT		
19.	MOVSW		
20.	INT 21h		

Question 4. What are the machine codes of following instructions?		**Point: 50**	**Get:**
No.	**Instruction**	**Machine languages/codes**	
1.	ADD CH, BL		

2.	SUB CH, CL	
3.	CMP BH,03h	
4.	DIV CH	
5.	MUL DL	
6.	CLI	
7.	PUSH DX	
8.	MOV AH,01h	
9.	IN AL, 00h	
10.	OUT F0h, AX	

Question 5. Please draw the block diagrams of Von Neuman architecture and Harvard architecture?	**Point: 30**	**Get:**

Please give details about them.

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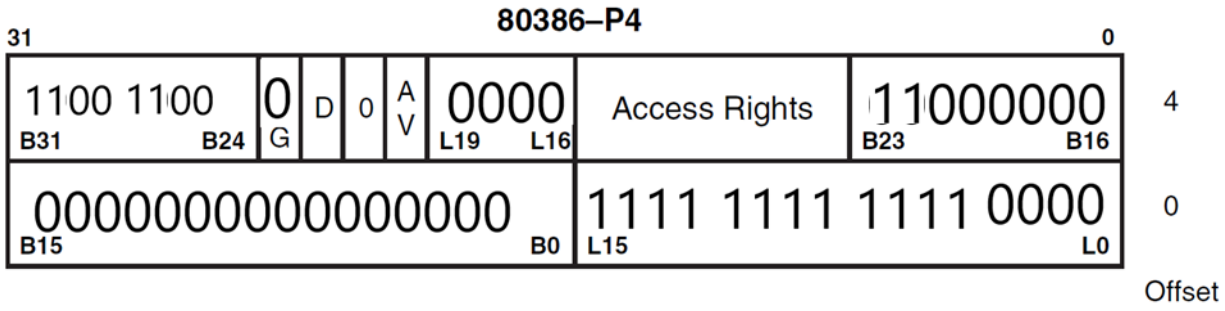
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Question 6. Please calculate physical addresses in the real mode for 8086/88?		**Point: 20**	**Get:**
1.	8567:49A0h		
2.	99A0:5567h		
3.	100F:00FFh		
4.	C0FF:D00F h		
5.	B00A:ACBAh		

Question 7. Give a descriptor of 80386 as below

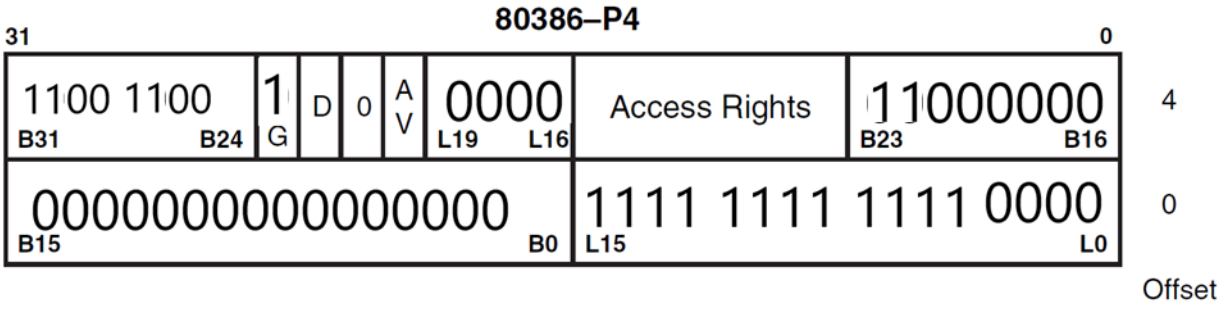


Please answer the below questions by filling the number or circle:

1. What is the range of address of the segment with above descriptor?			**Point: 10**	**Get:**
The range of address in hexadecimal is from …….. to.. ……				
2. The memory location at CCC0FFF0H is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. YES	B. NO	C. MAYBE		

3. The memory location at CCC00001H is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. YES	B. NO	C. MAYBE		

Question 8. Give a descriptor of 80386 as below



Please answer the below questions by filling the number or circle:

1. What is the range of address of the segment with above descriptor?			**Point: 10**	**Get:**
The range of address in hexadecimal is from …….. to.. ……				
2. The memory location at DCBF0FFF is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. YES	B. NO	C. MAYBE		
3. The memory location at D0000000H is belonged to the segment described by above descriptor?			**Point: 10**	**Get:**
A. YES	B. NO	C. MAYBE		
Question 9. In protected mode, what is the content of segment register?			**Point: 10**	**Get:**
A. Segment address	B. indicate the index of chosen descriptor	C. Both A) and B)		

Please give the brief explanation (if any)

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